Stochastic analysis and probability density function of viral infection model with Hattaf-Yousfi functional response

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Abstract

In this paper, we investigate a new stochastic viral infection model with Hattaf-Yousfi functional response. First, we prove that our stochastic model possesses a unique global positive solution. Next, by using the Lyapunov method, we determine some conditions for the extinction and the persistence of the infection. We also establish the existence of a unique stationary distribution for our stochastic model and dynamic behavior around the chronic infectious balance. Finally, numerical simulations are presented to support and to prove the applicability and efficiency of our theoretical results.

Keywords:

Viral infection, Stochastic differential equations, Extinction, Functional response, stationary distribution.

1 Introduction

In the last decades, many mathematicians and biologists have been interested in development and study of population dynamics of infectious diseases in order to explain and understand a behavior of epidemics and disease transmissions. There exists many examples of contagious viral diseases include the flu, HIV, Hepatitis B virus (HBV), herpes and human immunodeficiency virus (HIV). This virus is a complicated kinetic process that requires a mathematical models to understand the complexity between the body's responses and different conditions [15, 16] and references therein. The basic model of viral infection contains three variables, which are functions of time: the concentrations of uninfected cells (X_t), infected cells (Y_t), and free virus cells (V_t) which can be written by the ordinary differential equations (see [17]):

$$\begin{cases} dX_t &= (\lambda - \delta X_t - \beta X_t V_t)\, dt, \\ dY_t &= (-\gamma Y_t + \beta X_t V_t)\, dt, \\ dV_t &= (kY_t - \mu V_t)\, dt \end{cases} \tag{1.1}$$

in which the concentrations of uninfected cells (X_t) are produced at a constant rate, λ, die at the rate of δX_t, and become infected with the rate of $\beta X_t V_t$. Infected cells (Y_t) are produced at the rate of $\beta X_t V_t$ and die at the rate of γY_t. Free virus cells (V_t) are released from Y_t at the rate of kY_t and die at the rate of μV_t. In this case,we assume that all parameters λ, δ, β, γ, k and μ are positive constants.

Note that, many research suggest that the infection term is based on the mass-action principle, which means that there is a constant contact rate β between the concentrations of uninfected cells and free virus cells in (1.1). However, many experiments and research related to epidemic and micro-parasitic infections suggest that the infection rate is a nonlinear relationship [15, 13, 14] by replacing β with a continuous and differentiated function f which satisfied the following assumption:

$$(H_0): \quad f(0) = \beta, \quad f'(v) < 0, \quad \lim_{v \longrightarrow \infty} f(v) = 0. \tag{1.2}$$

Chenxi et al show that viral infection model (1.1) under some conditions and assumptions admits the unique positive solution and the disease-free equilibrium is globally asymptotically stable. In this paper, we consider the following model with specific nonlinear infection rate (see [12] for more details):

$$\begin{cases} dX_t &= (\lambda - \delta X_t - f(X_t, V_t)V_t)\, dt, \\ dY_t &= (-\gamma Y_t + f(X_t, V_t)V_t)\, dt, \\ dV_t &= (kY_t - \mu V_t)\, dt. \end{cases} \tag{1.3}$$

In the above model, the parameter λ is the recruitment rate of uninfected cells, δ is the death rates of uninfected cells, γ is the infected cells, μ is the free virus and k is the production rate of the free virus by an infected cell. The Hattaf-Yousfi functional response f is used to describe the incidence rate and its expression is given in [4, 5, 6], as follows

$$f(x, v) = \frac{\beta x}{\alpha_0 + \alpha_1 x + \alpha_2 v + \alpha_3 xv} \tag{1.4}$$

where $\alpha_0 > 0$, $\alpha_1, \alpha_2, \alpha_3 \geq 0$ and β is the infection coefficient. Many researchers are interested in the form of the function response f, see for instance [12, 5] and references therein. For example, if $\alpha_0 = 1$ and $\alpha_i = 0, i = 1, 2, 3$, f is called the mass-action or bilinear incidence function. If $\alpha_0 = 1$ and $\alpha_i = 0, i = 1, 3$, functional response f represents the saturation incidence rate.

Our interest in the topic of this work arose from the fact that the environmental fluctuations have great influence on all aspects of real life and there exist original paper which use and discuss the stochastic modeling of viral infection disease, see for instance, [4] which discus a version of stochastic differential equations with the unique intensity associated to free virus, but none of them has proposed a close-to-reality a stochastic model to understand a behavior of epidemics and disease transmissions. Therefore, the main goal of this paper is to carry out a stochastic modeling and mathematical analysis of viral infection model and predict whether the infection disappears or survives.

Connected to the above statements, we suggest to study the dynamics of a stochastic viral infection model with environmental noise proportional to X_t, Y_t and V_t which described by the following system:

$$\begin{cases} dX_t &= (\lambda - \delta X_t - f(X_t, V_t)V_t)\, dt - \sigma_1 X_t dW_t^1, \\ dY_t &= (\eta V_t - \gamma Y_t + f(X_t, V_t)V_t)\, dt - \sigma_2 Y_t dW_t^2, \\ dV_t &= (kY_t - (\mu + \eta)V_t)\, dt - \sigma_3 V_t dW_t^3 \end{cases} \tag{1.5}$$

with η the recover rate of infective such that $\gamma(\eta + \mu) > \eta k$ and (W_t^i), $1 \leq i \leq 3$ Standard Brownian motions which are independent with the respective intensities σ_i, $1 \leq i \leq 3$. When $\eta = 0$, $\sigma_1 = 0$, and $\sigma_2 = 0$, we retrieve the proposed model of viral infection cited in [4, Section 1]. Here, the authors prove that the stochastic model (1.3) admits a unique nonnegative solution on $\mathbb{R}_+^3$ with any given initial conditions. They also show the stability of the mathematical model and they establish the dynamical behavior around the chronic infection equilibrium $E_{eq}\left(x_{eq}, \frac{(\lambda - \delta x_{eq})(\mu+\eta)}{\gamma(\mu+\eta) - \eta k}, \frac{k(\lambda - \delta x_{eq})}{\gamma(\mu+\eta) - \eta k}\right)$ where x_{eq} is the unique solution of $f\left(x, \frac{k(\lambda - \delta x)}{\gamma(\mu+\eta) - \eta k}\right) = \frac{\gamma(\eta+\mu) - \eta k}{k}$ on $(0, +\infty)$.

The outline of this paper is structured as follows. In section 2 we show the well posedness of the stochastic model (1.5) by proving the existence of unique positive solution. In Section 3, we establish

the extinction of the infection in terms of the reproduction number R_0, and we study the persistence and the dynamical behavior around the chronic infection in mean result under some appropriate conditions. The existence of an ergodic stationary distribution to stochastic viral infection model is analyzed in Section 4. Finally, in Section 5, we present some numerical simulations to validate our theoretical results and we draw conclusion.

2 Existence and uniqueness of the global positive solution

In this section, we show that system (1.5) has a unique nonnegative solution for any initial conditions on $\mathbb{R}^3_+$.

Theorem 1 *For any given initial value* $(X_0, Y_0, V_0) \in \mathbb{R}^3_+$*, there exists a unique local solution* $(X_t, Y_t, V_t) \in \mathbb{R}^3_+$ *to stochastic model* (1.5) *defined on* $[0, +\infty[$ *with probability one.*

Proof.

In order to show the existence of a unique nonnegative solution, we adopt the strategy from [1]. Since the system (1.5) has a locally Lipschitz coefficients of linear growth, there exists a unique solution (X_t, Y_t, V_t) for all $0 \leq t < \tau_e$ where τ_e is the explosion time. Now, we need to prove that $\tau = \infty$ almost surely. Assume that this statement is false, then there is a constant $T > 0$ such that $\mathbb{P}(\tau < T) > 0$. To do this, we set introduce the C^2-function $U : \mathbb{R}^3_+ \longrightarrow \mathbb{R}_+$ as follows

$$U(x, y, v) = a\varphi\left(\frac{\delta}{\lambda}x\right) + b\varphi(y) + c\varphi(v) \tag{2.1}$$

where $\varphi(x) = x - 1 - \ln(x)$ and the positive parameters $(a,\, b,\, c)$ will be precised later. By applying Itô's formula to U, we have

$$\begin{aligned} dU(X_t, Y_t, V_t) =& \mathcal{L}U(X_t, Y_t, V_t)dt + a\left(1 - \frac{\delta}{\lambda X_t}\right)\sigma_1 X_t dW_t^1 \\ &+ b\left(1 - \frac{1}{Y_t}\right)\sigma_2 Y_t dW_t^2 + c\left(1 - \frac{1}{V_t}\right)\sigma_3 V_t dW_t^3 \end{aligned} \tag{2.2}$$

where $\mathcal{L}$ is the operator defined by

$$\begin{aligned} \mathcal{L}U(X_t, Y_t, V_t) =& a\left(1 - \frac{\delta}{\lambda X_t}\right)(\lambda - \delta X_t - f(X_t, V_t)V_t) \\ &+ b\left(1 - \frac{1}{Y_t}\right)(\eta V_t - \gamma Y_t + f(X_t, V_t)V_t) \\ &+ c\left(1 - \frac{1}{V_t}\right)(kY_t - (\mu + \eta)V_t) + \frac{1}{2}\left(a\sigma_1^2 + b\sigma_2^2 + c\sigma_3^2\right). \end{aligned} \tag{2.3}$$

By arranging each term, one has

$$\begin{aligned} \mathcal{L}U(X_t, Y_t, V_t) =& -\frac{a\lambda}{X_t}\left(X_t - \frac{\delta}{\lambda}\right)^2 - \left(\frac{a}{Y_t}f(X_t, V_t)V_t + b\eta\frac{V_t}{Y_t} + ck\frac{Y_t}{V_t}\right) \\ &+ (b - a)f(X_t, V_t)V_t + (b\eta - (\mu + \eta)c)V_t + (kc - \gamma b)Y_t \\ &+ a\left(\frac{V_t}{X_t}f(X_t, V_t) + \frac{\sigma_1^2}{2}\right) + b\left(\gamma + \frac{\sigma_2^2}{2}\right) + c\left(\mu + \eta + \frac{\sigma_3^2}{2}\right). \end{aligned} \tag{2.4}$$

Choosing $a = \dfrac{\alpha_0\lambda}{\alpha_0\delta+\alpha_1\lambda}$, $b = \dfrac{\alpha_0\lambda}{\alpha_0\delta+\alpha_1\lambda}$, $c = \min\left(\frac{\eta}{\mu+\eta}, \frac{\gamma}{k}\right) \times \dfrac{\alpha_0\lambda}{\alpha_0\delta+\alpha_1\lambda}$ and using $f(X_t, V_t) \leq \frac{\beta X_t}{\alpha_2 V_t}$, one gets

$$\mathcal{L}U(X_t, Y_t, V_t) \leq C \tag{2.5}$$

where $C = \dfrac{\alpha_0\lambda}{(\alpha_0\delta+\alpha_1\lambda)}\left[\frac{\beta+\alpha_2\gamma}{\alpha_2} + \frac{\sigma_1^2+\sigma_2^2}{2} + \min\left(\frac{\eta}{\mu+\eta}, \frac{\gamma}{k}\right)\left(\mu+\eta+\frac{\sigma_3^2}{2}\right)\right]$.

The rest part of the proof is analogous to the one given in [**?**, Theorem 1.26]. But, for the convenience of lecturer, we will give it here. By injecting (2.5) in (2.2), we deduce

$$dU(X_t, Y_t, V_t) \leq Cdt - \left(1 - \frac{a}{X_t}\right)\sigma_1 dW_t^1 - \left(1 - \frac{1}{Y_t}\right)\sigma_2 dW_t^2 - \left(1 - \frac{1}{V_t}\right)\sigma_3 dW_t^3. \tag{2.6}$$

By integrating both sides of (2.6) between 0 and t, we can get

$$U(X_t, Y_t, V_t) \leq U(X_0, Y_0, V_0) + Ct + M_t \tag{2.7}$$

where

$$M_t = -\int_0^t \left(1 - \frac{a}{X_s}\right)\sigma_1 dW_s^1 - \int_0^t \left(1 - \frac{1}{Y_s}\right)\sigma_2 dW_s^2 - \int_0^t \left(1 - \frac{1}{V_s}\right)\sigma_3 dW_s^3. \tag{2.8}$$

Noticing some components of the vector solution (X_τ, Y_τ, V_τ) equal to 0. Then $\lim\limits_{t\longrightarrow\tau} U(X_t, Y_t, V_t) = +\infty$. Passing to the limit when t tend to τ in (2.7) leads to the contradiction

$$+\infty \leq U(X_0, Y_0, V_0) + C\tau + M_\tau < +\infty. \tag{2.9}$$

Hence $\tau = +\infty$ almost surely, which means that $(X_t, Y_t, V_t) \in \mathbb{R}_+^3$ for all $t \geq 0$.

3 Extinction of the viral infection

In this paragraph, we establish sufficient conditions for the extinction of the infection in the stochastic model (1.5). Let be introduced the following reproduction number $\mathcal{R}_0$:

$$\mathcal{R}^0 = \frac{k\eta}{(\mu+\eta)\gamma}. \tag{3.1}$$

Then, the following theorem is hold.

Theorem 2 *Let (X_t, Y_t, V_t) be the solution to stochastic model (1.5) for any initial condition $(X_0, Y_0, V_0) \in \mathbb{R}_+^3$. Suppose that $\mathcal{R}^0 < 1$ and*

$$\min(\gamma, \mu+\eta)\left(\sqrt{R_0} - 1\right) + \frac{(2\delta+\sigma_1^2)(\mu+\eta)\beta\sqrt{R_0}}{2\lambda\alpha_0\gamma} < 0.$$

Then, the disease dies out exponentially with probability one. Moreover, the distribution of $S(t)$ converges weakly to the measure which has the density

Proof.

The proof is analogous to the one given in [18, page 5], and it is based on the ergodic property and the limit sup of Brownian motion. We first interest in the ergodic property by using the same strategy in [3, Theorem 1.16, Chapter I]. Let $(S_t)_{t\geq 0}$ be solution to the following stochastic differential equation

$$dS_t = (\lambda - \delta S_t)dt + \sigma_1 S_t dW_t^1, S_0 = X_0 > 0. \tag{3.2}$$

We consider

$$q(x) = \exp\left(-2\int_1^x \frac{\lambda - \delta s}{\sigma_1^2 s^2} ds\right). \tag{3.3}$$

Then

$$q(x) = \exp\left(-\frac{2\lambda}{\sigma_1^2} + \frac{2\lambda}{\sigma_1^2 x} + \frac{2\delta}{\sigma_1^2}\ln x\right) = e^{-\frac{2\lambda}{\sigma_1^2}} e^{\frac{2\lambda}{\sigma_1^2 x}} x^{\frac{2\delta}{\sigma_1^2}}. \tag{3.4}$$

Form (3.4), we can verify easily that

$$\int_0^1 q(x)dx = +\infty, \quad \int_1^\infty q(x)dx = +\infty, \quad \int_0^\infty \frac{q(x)^{-1}}{\sigma_1^2 x^2} dx < +\infty. \tag{3.5}$$

Thus, by using the theorem 1.16 of law of the large numbers in [3, Chapter 1, Section 1.2], SDE (3.2) has the ergodic property with the density of its invariant law given by the following equation

$$p(x) = \left(\frac{2\lambda}{\sigma_1^2}\right)^{\frac{2\delta}{\sigma_1^2}+1} \left[\Gamma\left(\frac{2\delta}{\sigma_1^2}+1\right)\right]^{-1} x^{-\frac{2\delta}{\sigma_1^2}-2} e^{-\frac{2\lambda}{\sigma_1^2 x}}. \tag{3.6}$$

where Γ is a gamma function. Now, we consider the nonnegative C^2-function

$$U(Y_t, V_t) = \frac{w_1}{\gamma} Y_t + \frac{w_2}{\mu+\eta} V_t \tag{3.7}$$

where w_1 and w_2 are positive constants to be precised later. Using Itô's formula for $\ln U$, one gets

$$\begin{aligned} d\ln U =& \frac{1}{U}\left(\frac{w_1}{\gamma}\left(\eta V_t - \gamma Y_t + f(X_t, V_t)V_t\right) + \frac{w_2}{\mu+\eta}\left(kY_t - (\mu+\eta)V_t\right)\right) dt \\ &- \frac{1}{U^2}\left(\frac{w_1\sigma_2^2}{\gamma} Y_t^2 + \frac{w_2\sigma_3^2}{\mu+\eta} V_t^2\right) dt + \frac{1}{U}\left(\frac{w_1\sigma_2}{\gamma} Y_t dW_t^2 + \frac{w_2\sigma_3}{\mu+\eta} V_t dW_t^3\right) \\ =& \mathcal{L}U dt + \frac{1}{U}\left(\frac{w_1\sigma_2}{\gamma} Y_t dW_t^2 + \frac{w_2\sigma_3}{\mu+\eta} V_t dW_t^3\right) \end{aligned} \tag{3.8}$$

where

$$\begin{aligned} \mathcal{L}U =& \frac{1}{U}\left\{\left(\frac{w_1\eta}{\gamma} - w_2\right) V_t + \left(\frac{kw_2}{\mu+\eta} - w_1\right) Y_t + f(X_t, V_t)V_t\right\} \\ &- \frac{1}{U^2}\left(\frac{w_1\sigma_2^2}{\gamma} Y_t^2 + \frac{w_2\sigma_3^2}{\mu+\eta} V_t^2\right) \\ \leq& \frac{1}{U}\left\{\left(\frac{w_1\eta}{\gamma} - w_2\right) V_t + \left(\frac{kw_2}{\mu+\eta} - w_1\right) Y_t + \frac{w_1}{\gamma} f(X_t, V_t)V_t\right\} \\ =& \frac{1}{U}\left(w_1\, w_2\right)\left(M(Y_t\, V_t)^T - (Y_t\, V_t)\right) + \frac{w_1}{\gamma U} f(X_t, V_t)V_t \\ \leq& \frac{1}{U}\left(w_1\, w_2\right)\left(M(Y_t\, V_t)^T - (Y_t\, V_t)\right) + \frac{w_1(\mu+\eta)\beta}{\alpha_0\gamma} X_t \end{aligned} \tag{3.9}$$

where the matrix M is given as follows

$$M = \begin{pmatrix} 0 & \frac{\eta}{\gamma} \\ \dfrac{k}{\mu+\eta} & 0 \end{pmatrix} \tag{3.10}$$

and it has the greatest eigenvalue $\sqrt{R_0} = \sqrt{\frac{k\eta}{(\mu+\eta)\gamma}}$. By choosing $(w_1, w_2) = \left(\sqrt{R_0}, \frac{\eta}{\gamma}\right)$, one has

$$(w_1\, w_2)\, M = \sqrt{R_0}\,(w_1\, w_2)\,. \tag{3.11}$$

By putting together (3.9), (3.11) and (3.7), one gets

$$\begin{aligned}\mathcal{L}U &\leq \frac{1}{U}\left(\sqrt{R_0}-1\right)(w_1 Y_t + w_2 V_t) + \frac{w_1(\mu+\eta)\beta}{\alpha_0\gamma} X_t \\ &\leq \min(\gamma, \mu+\eta)\left(\sqrt{R_0}-1\right) + \frac{(\mu+\eta)\beta\sqrt{R_0}}{\alpha_0\gamma} X_t \\ &\leq \min(\gamma, \mu+\eta)\left(\sqrt{R_0}-1\right) + \frac{(\mu+\eta)\beta\sqrt{R_0}}{\alpha_0\gamma} S_t. \end{aligned} \tag{3.12}$$

Hence, the following inequality holds

$$\begin{aligned} d\ln U \leq &\min(\gamma, \mu+\eta)\left(\sqrt{R_0}-1\right) + \frac{(\mu+\eta)\beta\sqrt{R_0}}{\alpha_0\gamma} S_t \\ &- \frac{\sqrt{R_0}}{\gamma U}\left(\sigma_2 Y_t dW_t^2 + \sqrt{R_0}\sigma_3 V_t dW_t^3\right). \end{aligned} \tag{3.13}$$

Integrating both sides of the last inequality from 0 to t and dividing by t, we arrive at the following statement

$$\begin{aligned} \frac{\ln U(Y_t, V_t)}{t} \leq &\frac{\ln U(Y_0, V_0)}{t} + \min(\gamma, \mu+\eta)\left(\sqrt{R_0}-1\right) \\ &+ \frac{(\mu+\eta)\beta\sqrt{R_0}}{\alpha_0\gamma}\frac{1}{t}\int_0^t S_u du - \frac{M_t}{t} \end{aligned} \tag{3.14}$$

where $M_t = \int_0^t \left(\frac{\sqrt{R_0}}{\gamma U}\left(\sigma_2 Y_t dW_t^2 + \sqrt{R_0}\sigma_3 V_t dW_t^3\right)\right) dt$ is local martingale with finite quadratic variation. By using the strong law of large numbers for local martingales, we deduce

$$\limsup_{t\longrightarrow+\infty} \frac{M_t}{t} = 0 \quad a.s. \tag{3.15}$$

In order to estimate , we apply the ergodic property of the solution to system (3.2) and we use $\int_0^\infty xp(x)dx < +\infty$. Then

$$\lim_{t\longrightarrow+\infty} \frac{1}{t}\int_0^t S_u du = \int_0^\infty xp(x)dx = \frac{2\delta+\sigma_1^2}{2\lambda}. \tag{3.16}$$

By passing to the limit when t tends to infinity, we arrive at the desired result

$$\limsup_{t\longrightarrow+\infty} \frac{\ln U(Y_t, V_t)}{t} \leq \min(\gamma, \mu+\eta)\left(\sqrt{R_0}-1\right) + \frac{(2\delta+\sigma_1^2)(\mu+\eta)\beta\sqrt{R_0}}{2\lambda\alpha_0\gamma} < 0\, a.s. \tag{3.17}$$

Moreover, we get

$$\limsup_{t\longrightarrow+\infty} \frac{V_t}{t} \leq 0, \quad \limsup_{t\longrightarrow+\infty} \frac{Y_t}{t} \leq 0 \quad a.s. \tag{3.18}$$

We conclude from (3.18) that V_t tends to 0 exponentially with probability one. Then, the proof is complete.

3.1 Persistence in mean of the infection

In order to know more virus behaviors, we are now interested in the persistence of diseases. Since, it provides the conditions under which the diseases are prevalent. For the sake of simplicity, we introduce the following quantity

$$\mathcal{R}_0^s = \frac{2k\eta\delta}{\delta\left(\gamma+\frac{\sigma_2^2}{2}\right)\left(\mu+\eta+\frac{\sigma_3^2}{2}\right)+k\eta\left(\delta+\frac{\sigma_1^2}{2}\right)}. \tag{3.19}$$

Theorem 3 *If $R_0^s > 1$, then for any given initial value $(X_0, Y_0, V_0) \in \mathbb{R}_+^3$, the corresponding solution to (3) satisfies*

$$\liminf_{t\longrightarrow+\infty} \frac{1}{t}\int_0^t V_s ds \geq \frac{2\alpha_0}{\beta}\left(\delta+\frac{\sigma_1^2}{2}\right)\left(1-\frac{1}{\mathcal{R}_0^s}\right) \quad a.s. \tag{3.20}$$

In other words, the viral infection will be permanent.

Proof.

Let $(X_0, Y_0, V_0) \in \mathbb{R}_+^3$ and let us define the function U such that

$$U(X_t, Y_t, V_t) = -w_1 \ln X_t - w_2 \ln Y_t - w_3 \ln V_t + X_t \tag{3.21}$$

where w_1, w_2 and w_3 are positive constants to be determined. It is easy to verify that the function U is continuous and goes to infinity as (X_t, Y_t, V_t) goes to the boundary of $\mathbb{R}_+^3$. Thus it must has a lower bound M. Therefore, we define the nonnegative function $\tilde{U} = U(X_t, Y_t, V_t) - M$. By applying Itô's formula to $\tilde{U}$, we arrive at

$$d\tilde{U} = \mathcal{L}\tilde{U}dt - (w_1 + X_t)\sigma_1 dW_t^1 - w_2\sigma_2 dW_t^2 - w_3\sigma_3 dW_t^3 \tag{3.22}$$

where

$$\begin{aligned}\mathcal{L}\tilde{U} = &-\frac{w_1\lambda}{X_t} - \delta X_t - w_2\eta\frac{V_t}{Y_t} - kw_3\frac{Y_t}{V_t} - \left(\frac{w_2}{Y_t}+w_4\right)V_t f(X_t, V_t) + \lambda \\ &+ w_1 f(X_t, V_t)\frac{V_t}{X_t} + w_1\left(\delta+\frac{\sigma_1^2}{2}\right) + w_2\left(\gamma+\frac{\sigma_2^2}{2}\right) + w_3\left(\mu+\eta+\frac{\sigma_3^2}{2}\right).\end{aligned} \tag{3.23}$$

Using the fact that $f(X_t, V_t) \leq \dfrac{\beta X_t}{\alpha_0}$ and applying the arithmetic-geometric inequality, we arrive at

$$\begin{aligned}\mathcal{L}\tilde{U} \leq &-2\sqrt{w_1\delta\lambda} - 2\sqrt{w_2 w_3 k\eta} + \frac{w_1\beta}{\alpha_0}V_t \\ &+ w_1\left(\delta+\frac{\sigma_1^2}{2}\right) + w_2\left(\gamma+\frac{\sigma_2^2}{2}\right) + w_3\left(\mu+\eta+\frac{\sigma_3^2}{2}\right) + \lambda.\end{aligned} \tag{3.24}$$

Choosing $w_1 = \frac{\lambda I^2}{\delta}$, $w_2 = \lambda I\left(\delta+\frac{\sigma_2^2}{2}\right)^{-1}$ and $w_3 = \frac{\lambda I}{k\eta}\left(\gamma+\frac{\sigma_2^2}{2}\right)$ where $I = \delta\left(\delta+\frac{\sigma_1^2}{2}\right)^{-1}$, one gets

$$\begin{aligned}\mathcal{L}\tilde{U} &\leq -2\lambda I\left(1-\frac{1}{2k\eta}\left(\gamma+\frac{\sigma_2^2}{2}\right)\left(\mu+\eta+\frac{\sigma_3^2}{2}\right)-\frac{1}{2I}\right) + \frac{\beta\lambda I^2}{\alpha_0\delta}V_t \\ &\leq -2\lambda I\left(1-\frac{1}{\mathcal{R}_0^s}\right) + \frac{\beta\lambda I^2}{\alpha_0\delta}V_t.\end{aligned} \tag{3.25}$$

Injecting (3.25) in (3.26), we obtain

$$d\tilde{U} \leq -2\lambda I\left(1-\frac{1}{\mathcal{R}_0^s}\right)dt+\frac{\beta\lambda I^2}{\alpha_0\delta}V_t dt-(w_1+X_t)\sigma_1 dW_t^1-w_2\sigma_2 dW_t^2-w_3\sigma_3 dW_t^3. \tag{3.26}$$

Integrating both sides of the last inequality from 0 to t and dividing by t, we arrive at the following statement

$$0 \leq \frac{\tilde{U}\left(X_t,Y_t,V_t\right)}{t} \leq \frac{\tilde{U}\left(X_0,Y_0,V_0\right)}{t}-2\lambda I\left(1-\frac{1}{\mathcal{R}_0^s}\right)+\frac{\beta\lambda I^2}{\alpha_0\delta}\frac{1}{t}\int_0^t V_s ds-\frac{M_t}{t} \tag{3.27}$$

where $M_t = w_1\sigma_1 W_t^1+\sigma_1\int_0^t X_s dW_s^1+w_2\sigma_2 W_t^2+w_3\sigma_3 W_t^3$ is a continuous local martingale. By using the theorem 1.16 of law of the large numbers for martingales cited in [3, Chapter 1, Section 1.2], we obtain

$$\lim_{t\to+\infty}\frac{M_t}{t}=0. \tag{3.28}$$

Therefore

$$\liminf_{t\to+\infty}\frac{1}{t}\int_0^t V_s ds \geq \frac{2\alpha_0\delta}{\beta I}\left(1-\frac{1}{\mathcal{R}_0^s}\right) \quad a.s. \tag{3.29}$$

By using $I=\delta\left(\delta+\frac{\sigma_1^2}{2}\right)^{-1}$, we arrive at the desired result (3.20). The proof is completed.

3.2 Dynamics around the chronic infection equilibrium

Also, the chronic infection equilibrium E^* is not a steady state of system (2). For this reason, we will discuss the dynamics around E^* when the noise is sufficiently small. Let be introduced

Theorem 4 *Let (X_t,Y_t,V_t) be the solution of system with initial value $(X_0,Y_0,V_0)\in\mathbb{R}_+^3$. Assume that $\sigma_1^2<\frac{\delta}{4}$, $\sigma_2^2<\frac{\gamma}{4}$, $\sigma_3^2\leq\frac{\mu+\eta}{2}$ and*

$$\begin{aligned} h=&\frac{2(\delta-4\sigma_1^2)+\delta}{2(\delta-4\sigma_1^2)(4\sigma_2^2+\gamma)+(2+\frac{\gamma}{\delta})}+\frac{2\lambda\gamma}{\delta\eta}\left(1+\frac{\alpha_1 x_{eq}+\alpha_3 x_{eq}v_{eq}}{\alpha_0+\alpha_2 v_{eq}}\right)\\ &-\frac{\gamma(\mu+\eta-2\sigma_3^2)(\delta-4\sigma_1^2)}{2k^2\eta^2(4\sigma_2^2+\delta)+2k^2\eta^2(2+\frac{\gamma}{\delta})}<0. \end{aligned}$$

Then

$$\limsup_{t\longrightarrow+\infty}\frac{1}{t}\int_0^t \mathbb{E}\left(|X_s-X_{eq}|^2+|Y_s-Y_{eq}|^2+|V_s-V_{eq}|^2\right)ds \leq \frac{A_{max}}{A_{min}}\left(\eta+\Sigma\right) \tag{3.30}$$

with

$$\begin{aligned} \Sigma&=\sigma_1^2\left(x_{eq}+\sqrt{\frac{c_1}{c_2}\frac{\alpha_0+\alpha_1 v_{eq}}{4\beta}}\right)^2+\sigma_2^2\left(y_{eq}+\frac{c_1}{4c_2}\right)^2+\sigma_3^2\left(v_{eq}+\frac{c_1}{2kc_3}\right)^2,\\ c_1&=\frac{\lambda}{\delta^2}\left(1+\frac{\alpha_1 x_{eq}+\alpha_3 x_{eq}v_{eq}}{\alpha_0+\alpha_2 v_{eq}}\right),\\ c_2&=\frac{\delta-4\sigma_1^2}{\delta(4\sigma_2^2+\delta)+\gamma(2\delta+\gamma)}\\ c_3&=\frac{\mu+\eta}{2k^2}\frac{\delta-4\sigma_1^2}{\delta(4\sigma_2^2+\delta)+\gamma(2\delta+\gamma)}\\ A_{min}&=\min\left(\frac{1}{2},\frac{\gamma c_2}{8},\frac{h\eta^2}{\delta\gamma}\right)\\ A_{max}&=\max\left(\frac{v_{eq}+2\gamma v_{eq}y_{eq}(1+v_{eq})}{\gamma y_{eq}}c_1,c_1,2c_2,c_3\right), \end{aligned}$$

Proof.

Let $(X_0, Y_0, V_0) \in \mathbb{R}_+^3$ and let us define the function U such that

$$U(X_t, Y_t, V_t) = c_1 U_1(X_t, Y_t, V_t) + c_2 U_2(X_t, Y_t) + c_3 U_3(V_t) \tag{3.31}$$

where

$$\begin{aligned}
U_1(x, y, v) &= x - \frac{\lambda}{\delta} - \int_{\frac{\lambda}{\delta}}^{x} \frac{f(x_{eq}, v_{eq})}{f(s, v_{eq})} ds + y_{eq}\varphi\left(\frac{y}{y_{eq}}\right) + \frac{\gamma v_{eq}}{k}\varphi\left(\frac{v}{v_{eq}}\right), \\
U_2(x, y) &= \frac{1}{2}\left(x + y - (x_{eq} + y_{eq})\right)^2, \\
U_3(v) &= \frac{1}{2}\left(v - v_{eq}\right)^2, \\
\varphi(x) &= x - 1 - \ln(x)
\end{aligned}$$

and (c_1, c_2, c_3) are the positive constants to be precised. Here, the first tree terms in the Lyapunov function U_1 add up to a non-negative quantity φ. Indeed, if $x \geq \frac{\lambda}{\delta}$ then

$$\int_{\frac{\lambda}{\delta}}^{x} \frac{f(x_{eq}, v_{eq})}{f(s, v_{eq})} ds \leq \int_{\frac{\lambda}{\delta}}^{x} \frac{f(x_{eq}, v_{eq})}{f(x_{eq}, v_{eq})} ds = x - \frac{\lambda}{\delta} \tag{3.32}$$

since $f(x, v_{eq})$ is non-decreasing. If $x \leq \frac{\lambda}{\delta}$, we apply the similar reasoning. For this reason, we interest in the sequel to the first case $X_t \geq \frac{\lambda}{\delta}$. By using the famous Itô's formula, we arrive at

$$dU(X_t, Y_t, V_t) = \sum_{i=1}^{3} c_i \mathcal{L}U_i dt + dM_t. \tag{3.33}$$

where

$$\begin{aligned}
\mathcal{L}U_1 &= \left(1 - \frac{f(x_{eq}, v_{eq})}{f(X_t, v_{eq})}\right)(\lambda - \delta X_t) + \frac{f(x_{eq}, v_{eq})}{f(X_t, v_{eq})} f(X_t, V_t) V_t \\
&\quad - y_{eq}\frac{V_t}{Y_t} f(X_t, V_t) - \eta y_{eq}\frac{V_t}{Y_t} - \frac{\gamma v_{eq}}{k}\frac{Y_t}{V_t} - \frac{\gamma(\mu + \eta)}{k} V_t + \eta V_t \\
&\quad + \gamma y_{eq} + \frac{\gamma(\mu + \eta)}{k} v_{eq} + \frac{\alpha_0 + \alpha_1 v_{eq}}{\beta}\sigma_1^2 + \sigma_2^2 y_{eq} + \frac{\gamma \sigma_3^2}{k} v_{eq}, \\
\mathcal{L}U_2 &= (X_t - x_{eq} + Y_t - y_{eq})(\eta V_t - \gamma Y_t + \lambda - \delta X_t) \\
&\quad + \frac{1}{2}(\sigma_1 X_t + \sigma_2 Y_t)^2, \\
\mathcal{L}U_3 &= (V_t - v_{eq})(kY_t - (\mu + \eta)V_t) + \frac{\sigma_3^2 V_t^2}{2}, \\
dM_t &= -\sigma_1\left[c_1\left(1 - \frac{f(x_{eq}, v_{eq})}{f(x, v_{eq})}\right) + c_2(X_t - x_{eq} + Y_t - y_{eq})\right] X_t dW_t^1 \\
&\quad - \sigma_2\left[(c_1 + c_2)(Y_t - y_{eq}) + c_2(X_t - x_{eq})\right] Y_t dW_t^2 \\
&\quad - \sigma_3(V_t - v_{eq})\left(c_1\frac{\gamma}{k} + c_3 V_t\right) dW_t^3.
\end{aligned}$$

In order to bound (3.30), we will be estimated each term $\mathcal{L}U_1$, $\mathcal{L}U_1$ and $\mathcal{L}U_3$. First, we will interest to estimate $\mathcal{L}U_1$. By using the equalities $\lambda = \delta x_{eq} + f(x_{eq}, v_{eq})v_{eq}$, $\gamma y_{eq} = \eta v_{eq} + f(x_{eq}, v_{eq})v_{eq}$ and

$y_{eq} = \frac{\mu+\eta}{k} v_{eq}$, we arrive at

$$\begin{aligned}\mathcal{L}U_1 = \delta\left(1 - \frac{f(x_{eq}, v_{eq})}{f(X_t, v_{eq})}\right)(x_{eq} - X_t) + 3f(x_{eq}, v_{eq})v_{eq} - \frac{f^2(x_{eq}, v_{eq})v_{eq}}{f(X_t, v_{eq})} \\ - y_{eq}\frac{V_t}{Y_t}f(X_t, V_t) - \eta y_{eq}\frac{V_t}{Y_t} - \frac{\gamma v_{eq}}{k}\frac{Y_t}{V_t} - \gamma\frac{y_{eq}}{v_{eq}}V_t + \frac{f(x_{eq}, v_{eq})}{f(X_t, v_{eq})}f(X_t, V_t)V_t \\ + \eta V_t + 2\eta v_{eq} + \frac{\alpha_0 + \alpha_1 v_{eq}}{2\beta}\sigma_1^2 + \frac{\sigma_2^2}{2}y_{eq} + \frac{\gamma\sigma_3^2}{2k}v_{eq}.\end{aligned} \tag{3.34}$$

By injecting the expression of f in the last equation and using $X_t \leq \frac{\lambda}{\delta}$, we obtain

$$\begin{aligned}\mathcal{L}U_1 = -\frac{\delta^2(\alpha_0 + \alpha_2 v_{eq})(X_t - x_{eq})^2}{\lambda(\alpha_0 + \alpha_1 x_{eq} + \alpha_2 v_{eq} + \alpha_3 x_{eq} v_{eq})} + \eta V_t^2 - \eta V_t\left(2 + \frac{y_{eq}}{Y_t}\right) \\ - \frac{f(x_{eq}, v_{eq})(\alpha_0 + \alpha_1 x_{eq})(\alpha_2 + \alpha_3 x_{eq})(V_t - v_{eq})2}{(\alpha_0 + \alpha_1 X_t + \alpha_0 + \alpha_3 X_t V_t)(\alpha_0 + \alpha_1 X_t + \alpha_2 v_{eq} + \alpha_3 X_t v_{eq})} \\ + f(x_{eq}, v_{eq})v_{eq}\left(4 - \frac{f(x_{eq}, v_{eq})}{f(X_t, v_{eq})} - \frac{y_{eq}V_t}{v_{eq}Y_t}\frac{f(X_t, V_t)}{f(x_{eq}, v_{eq})} - \frac{v_{eq}Y_t}{y_{eq}V_t} - \frac{f(X_t, v_{eq})}{f(X_t, V_t)}\right) \\ - \frac{v_{eq}}{k}\left(\frac{kv_{eq}}{y_{eq}} + (1-k)\gamma\right)\frac{Y_t}{V_t} + \eta V_t\left(3 - \frac{\gamma y_{eq}}{\eta v_{eq}} - \frac{\eta v_{eq}}{\gamma y_{eq}}\frac{1}{V_t} - V_t\right) \\ + \frac{\eta v_{eq}}{\gamma y_{eq}} + 2\eta v_{eq} + \frac{\alpha_0 + \alpha_1 v_{eq}}{2\beta}\sigma_1^2 + \frac{\sigma_2^2}{2}y_{eq} + \frac{\gamma\sigma_3^2}{2k}v_{eq}.\end{aligned} \tag{3.35}$$

By applying the arithmetic-geometric inequality, we obtain

$$\left(4 - \frac{f(x_{eq}, v_{eq})}{f(X_t, v_{eq})} - \frac{y_{eq}V_t}{v_{eq}Y_t}\frac{f(X_t, V_t)}{f(x_{eq}, v_{eq})} - \frac{v_{eq}Y_t}{y_{eq}V_t} - \frac{f(X_t, v_{eq})}{f(X_t, V_t)}\right) \leq 0 \tag{3.36}$$

$$\left(3 - \frac{\gamma y_{eq}}{\eta v_{eq}} - \frac{\eta v_{eq}}{\gamma y_{eq}}\frac{1}{V_t} - V_t\right) \leq 0. \tag{3.37}$$

By putting together (3.36), (3.37) and $k \leq 1$, we arrive at

$$\begin{aligned}\mathcal{L}U_1 \leq -\frac{\delta^2(\alpha_0 + \alpha_2 v_{eq})(X_t - x_{eq})^2}{\lambda(\alpha_0 + \alpha_1 x_{eq} + \alpha_2 v_{eq} + \alpha_3 x_{eq} v_{eq})} + 2\eta\left(V_t - v_{eq}\right)^2 \\ + \frac{\eta v_{eq}}{\gamma y_{eq}} + 2\eta v_{eq}(1 + v_{eq}) + \frac{\alpha_0 + \alpha_1 v_{eq}}{2\beta}\sigma_1^2 + \frac{\sigma_2^2}{2}y_{eq} + \frac{\gamma\sigma_3^2}{2k}v_{eq}.\end{aligned} \tag{3.38}$$

Secondly, by using $\lambda = \delta x_{eq} + f(x_{eq}, v_{eq})v_{eq}$, $\eta v_{eq} + f(x_{eq}, v_{eq})v_{eq} = \gamma y_{eq}$, we obtain

$$\begin{aligned}\mathcal{L}U_2 = -\delta(X_t - x_{eq})^2 - \gamma(Y_t - y_{eq})^2 - (\delta + \gamma)(X_t - x_{eq})(Y_t - y_{eq}) \\ + \eta(V_t - v_{eq})(X_t - x_{eq}) + \eta(V_t - v_{eq})(Y_t - y_{eq}) + \frac{1}{2}\left(\sigma_1 X_t + \sigma_2 Y_t\right)^2.\end{aligned}$$

Using Young inequality with $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$, we arrive at

$$\begin{aligned}\mathcal{L}U_2 &\leq -\delta(X_t - x_{eq})^2 - \gamma(Y_t - y_{eq})^2 - (\delta + \gamma)(X_t - x_{eq})(Y_t - y_{eq}) \\ &\quad + \eta(V_t - v_{eq})(X_t - x_{eq}) + \eta(V_t - v_{eq})(Y_t - y_{eq}) + \frac{1}{2}\left(\sigma_1 X_t + \sigma_2 Y_t\right)^2 \\ &\leq -\left(\delta - \frac{(\delta+\gamma)^2}{2\varepsilon_1} - \frac{\varepsilon_2}{2} - 2\sigma_1^2\right)(X_t - x_{eq})^2 - \left(\gamma - \frac{\varepsilon_1 + \varepsilon_3}{2} - 2\sigma_2^2\right)(Y_t - y_{eq})^2 \\ &\quad + \eta^2\left(\frac{1}{2\varepsilon_2} + \frac{1}{2\varepsilon_3}\right)(V_t - v_{eq})^2 + 2\left(\sigma_1^2 x_{eq}^2 + \sigma_2^2 y_{eq}^2\right).\end{aligned} \tag{3.39}$$

By combining $\sigma_1^2 < \frac{\delta}{4}$, $\sigma_2^2 < \frac{\gamma}{4}$, $\varepsilon_1 = \gamma - 4\sigma_2^2$, $\varepsilon_2 = \delta - 4\sigma_1^2$ and $\varepsilon_3 = \frac{\gamma}{2}$, we obtain

$$\begin{aligned}\mathcal{L}U_2 \leq & \frac{\delta(4\sigma_2^2+\delta)+\gamma(2\delta+\gamma)}{2(\delta-4\sigma_1^2)}(X_t-x_{eq})^2 - \frac{\gamma}{4}(Y_t-y_{eq})^2 \\ & + \frac{2\eta^2(\delta-4\sigma_1^2)+\gamma\eta^2}{2\gamma(\delta-4\sigma_1^2)}\left(V_t-v_{eq}\right)^2 + 2\left(\sigma_1^2 x_{eq}^2 + \sigma_2^2 y_{eq}^2\right). \end{aligned} \tag{3.40}$$

Moreover, we have

$$\mathcal{L}U_3 = k(V_t-v_{eq})(Y_t-y_{eq}) - (\mu+\eta)(V_t-v_{eq})^2 + \frac{\sigma_3^2 V_t^2}{2}. \tag{3.41}$$

Using $ky_{eq} = (\mu+\eta)v_{eq}$ and Young inequality, we obtain

$$\begin{aligned}\mathcal{L}U_3 &\leq \frac{k^2}{2(\mu+\eta)}(Y_t-y_{eq})^2 - \frac{\mu+\eta}{2}(V_t-v_{eq})^2 + \sigma_3^2(V_t-v_{eq})^2 + \sigma_3^2 v_{eq}^2 \\ &\leq \frac{k^2}{4(\mu+\eta)}(Y_t-y_{eq})^2 - \left(\frac{\mu+\eta}{2} - \sigma_3^2\right)(V_t-v_{eq})^2 + \sigma_3^2 v_{eq}^2. \end{aligned} \tag{3.42}$$

Therefore, by putting together (3.38), (3.40) and (3.42), we arrive at

$$\begin{aligned}\frac{dU(X_t,Y_t,V_t)}{dt} \leq & \left(c_2\frac{\delta(4\sigma_2^2+\delta)+\gamma(2\delta+\gamma)}{2(\delta-4\sigma_1^2)} - c_1\frac{\delta^2(\alpha_0+\alpha_2 v_{eq})}{\lambda(\alpha_0+\alpha_1 x_{eq}+\alpha_2 v_{eq}+\alpha_3 x_{eq}v_{eq})}\right) \\ & \times (X_t-x_{eq})^2 + \left(c_3\frac{k^2}{4(\mu+\eta)} - c_2\frac{\gamma}{4}\right)(Y_t-y_{eq})^2 \\ & + \left(c_2\frac{2\eta^2(\delta-4\sigma_1^2)+\gamma\eta^2}{2\gamma(\delta-4\sigma_1^2)} + 2c_1\eta - c_3\left(\frac{\mu+\eta}{2}-\sigma_3^2\right)\right)(V_t-v_{eq})^2 \\ & + \frac{c_1 v_{eq}\eta}{\gamma y_{eq}}\left(1+2\gamma y_{eq}(1+v_{eq})\right) + \sigma_1^2\left(c_1\frac{\alpha_0+\alpha_1 v_{eq}}{2\beta} + 2c_2 x_{eq}^2\right) \\ & + \sigma_2^2\left(2c_2 y_{eq}^2 + \frac{c_1 y_{eq}}{2}\right) + \sigma_3^2\left(c_3 v_{eq}^2 + \frac{c_1\gamma}{2k}v_{eq}\right) + \frac{dM_t}{dt}. \end{aligned} \tag{3.43}$$

Choosing $c_1 = \frac{\lambda}{\delta^2}\left(1+\frac{\alpha_1 x_{eq}+\alpha_3 x_{eq}v_{eq}}{\alpha_0+\alpha_2 v_{eq}}\right)$, $c_2 = \frac{\delta-4\sigma_1^2}{\delta(4\sigma_2^2+\delta)+\gamma(2\delta+\gamma)}$ and $c_3 = \frac{\mu+\eta}{2k^2}\frac{\delta-4\sigma_1^2}{\delta(4\sigma_2^2+\delta)+\gamma(2\delta+\gamma)}$, we have

$$\begin{aligned}\frac{dU(X_t,Y_t,V_t)}{dt} \leq & -\frac{1}{2}(X_t-x_{eq})^2 - \frac{\gamma c_2}{8}(Y_t-y_{eq})^2 - \frac{h\eta^2}{\delta\gamma}\left(V_t-v_{eq}\right)^2 \\ & + \frac{c_1 v_{eq}\eta}{\gamma y_{eq}}\left(1+2\gamma y_{eq}(1+v_{eq})\right) + \sigma_1^2\left(c_1\frac{\alpha_0+\alpha_1 v_{eq}}{2\beta} + 2c_2 x_{eq}^2\right) \\ & + \sigma_2^2\left(2c_2 y_{eq}^2 + \frac{c_1 y_{eq}}{2}\right) + \sigma_3^2\left(c_3 v_{eq}^2 + \frac{c_1\gamma}{2k}v_{eq}\right) + \frac{dM_t}{dt}. \end{aligned} \tag{3.44}$$

with

$$\begin{aligned} h = & \frac{2(\delta-4\sigma_1^2)+\delta}{2(\delta-4\sigma_1^2)(4\sigma_2^2+\gamma)+(2+\frac{\gamma}{\delta})} + \frac{2\lambda\gamma}{\delta\eta}\left(1+\frac{\alpha_1 x_{eq}+\alpha_3 x_{eq}v_{eq}}{\alpha_0+\alpha_2 v_{eq}}\right) \\ & - \frac{\gamma(\mu+\eta-2\sigma_3^2)(\delta-4\sigma_1^2)}{2k^2\eta^2(4\sigma_2^2+\delta)+2k^2\eta^2(2+\frac{\gamma}{\delta})} < 0. \end{aligned}$$

Taking $w = \frac{v_{eq}(1+2\gamma y_{eq}(1+v_{eq}))}{\gamma y_{eq}}$, choosing $A_{min} = \min\left(\frac{1}{2}, \frac{\gamma c_2}{8}, \frac{h\eta^2}{\delta\gamma}\right)$ and $A_{max} = \max\left(wc_1, c_1, 2c_2, c_3\right)$, we get

$$\begin{aligned}\frac{dU(X_t,Y_t,V_t)}{dt} &\leq -A_{min}\left((X_t - x_{eq})^2 + (Y_t - y_{eq})^2 + (V_t - v_{eq})^2\right)\\ &+ A_{max}\eta + A_{max}\sigma_1^2\left(x_{eq} + \sqrt{\frac{c_1}{c_2}\frac{\alpha_0 + \alpha_1 v_{eq}}{4\beta}}\right)^2\\ &+ A_{max}\sigma_2^2\left(y_{eq} + \frac{c_1}{4c_2}\right)^2 + A_{max}\sigma_3^2\left(v_{eq} + \frac{c_1}{2kc_3}\right)^2 + \frac{dM_t}{dt}.\end{aligned}$$

Multiplying both sides of the last inequality by dt, integrating between 0 and t and dividing by t, we arrive at

$$\begin{aligned}\frac{U(X_t,Y_t,V_t) - U(X_0,Y_0,V_0)}{t} &\leq -\frac{A_{min}}{t}\int_0^t \left((X_s - x_{eq})^2 + (Y_s - y_{eq})^2 + (V_s - v_{eq})^2\right) ds\\ &+ A_{max}\left[\eta + \sigma_1^2\left(x_{eq} + \sqrt{\frac{c_1}{c_2}\frac{\alpha_0 + \alpha_1 v_{eq}}{4\beta}}\right)^2\right.\\ &\left.+\sigma_2^2\left(y_{eq} + \frac{c_1}{4c_2}\right)^2 + \sigma_3^2\left(v_{eq} + \frac{c_1}{2kc_3}\right)^2\right] + \frac{1}{t}\int_0^t dM_s.\end{aligned}$$

Applying the theorem 1.16 of law of the large numbers for martingales cited in [3, Chapter 1, Section 1.2], to continuous local martingale $\frac{1}{t}\int_0^t dM_s$, we obtain

$$\lim_{t\to+\infty}\frac{1}{t}\int_0^t dM_s = 0. \tag{3.45}$$

Therefore, the following inequality

$$\limsup_{t\to+\infty}\frac{1}{t}\int_0^t \mathbb{E}\left((X_s - x_{eq})^2 + (Y_s - y_{eq})^2 + (V_s - v_{eq})^2\right) ds \leq \frac{A_{max}}{A_{min}}\left(\eta + \Sigma\right)$$

holds with

$$\Sigma = \sigma_1^2\left(x_{eq} + \sqrt{\frac{c_1}{c_2}\frac{\alpha_0 + \alpha_1 v_{eq}}{4\beta}}\right)^2 + \sigma_2^2\left(y_{eq} + \frac{c_1}{4c_2}\right)^2 + \sigma_3^2\left(v_{eq} + \frac{c_1}{2kc_3}\right)^2.$$

4 Stationary distribution and positive recurrence

In the previous sections, we have investigated both the extinction and the permanence of the disease under some conditions. In this section, we will show that the solution to stochastic model (1.5) has a unique stationary distribution by using the method described in [10, 11]. In order to announce the ergodic stationary distribution, we need recall the technical lemma (see [7] for more details and proof).

Let X_t be a regular homogeneous Markov process in $\mathbb{R}^n$ described by the following stochastic differential equation :

$$dX_t = g(X_t)dt + \sum_{m=1}^{k}\sigma_m(X_t)dW_t^m, \quad X_0 = x, \tag{4.1}$$

with the associated diffusion matrix A is defined by

$$A(x) = \left(a_{ij}(x)\right), \quad a_{ij}(x) = \sum_{m=1}^{k}\sigma_m^{(i)}(X_t)\sigma_m^{(j)}(X_t). \tag{4.2}$$

and the differential operator $\mathcal{L}$ associated to (4.1) is defined as follows

$$\mathcal{L}\cdot = g(x)^T\partial_x \cdot + \frac{1}{2}trace(A(x)\partial_{xx}\cdot). \tag{4.3}$$

Hypothesis: There exists a bounded domain $D \subset \mathbb{R}^n$ with regular boundary ∂D, having two properties that

(H_1) There exists a positive number N such that

$$\sum_{m=1}^{k} a_{ij}(x)\eta_i\eta_j \geq N|\eta|^2, \quad \eta \in \mathbb{R}^k, \quad x \in D.$$

(i.e. $\mathcal{L}U$ is uniformly elliptical in D, see [9, 8] for more details.)

(H_2) There exists a neighborhood $D^c := \mathbb{R}^n \setminus D$ and a nonnegative C^2-function $U : Dc \longrightarrow \mathbb{R}$ such that, for any $x \in D^c$, $\mathcal{L}U$ is negative (see [10]).

Remark 1 *We note that the diffusion matrix of* (1.5) *is* $A(X) = (\sigma_1 X)^2 + (\sigma_2 Y)^2 + (\sigma_3 V)^2$.

Lemma 1 *(Hasminskii [7]) Suppose that* (H_1) *and* (H_2) *hold, then the Markov process* X_t *has a unique ergodic stationary distribution* $\mu(\cdot)$. *Let* $h(\cdot)$ *be a* $\mathbb{R}^n$*-function integrable with respect to the measure* μ. *Then*

$$\mathbb{P}\left\{\lim_{T\to+\infty}\int_0^T h(X_t)dt = \int_{\mathbb{R}^n} h(x)\mu(dx)\right\} = 1. \tag{4.4}$$

Now we are in the position to announce our results of stationary distribution and positive recurrence.

Theorem 5 *For any given initial value* $(X_0, Y_0, V_0) \in \mathbb{R}^3_+$, *the corresponding solution* (X_t, Y_t, V_t) *to viral stochastic model* (1.5) *is positive recurrent and has a unique ergodic stationary distribution if the condition* $R_0^s > 1$ *holds.*

Proof.

In order to prove the result of the theorem 5, we need to verify two hypothesis (H_1) and (H_2). By using a remark **??**, it is easy to verify the first hypothesis (H_1). We only have to show the second hypothesis (H_2) is hold. To do this, we let introduce the C^2-function defined by

$$\overline{U}(X_t, Y_t, V_t) = MU(X_t, Y_t, V_t) + U_2(X_t, Y_t, V_t) + U_3(X_t, Y_t, V_t) \tag{4.5}$$

such that

$$\begin{aligned} U(X_t, Y_t, V_t) &= -w_1 \ln X_t - w_2 \ln Y_t - w_3 \ln V_t + X_t, \\ U_2(X_t, Y_t, V_t) &= -\ln X_t - \ln Y_t, \\ U_3(X_t, Y_t, V_t) &= \frac{1}{1+m}\left(X_t + Y_t + \frac{\gamma}{k}V_t\right)^{m+1} \end{aligned}$$

where (w_1, w_2, w_3) are given in the subsection 3.1, M and m are positive constants satisfying some conditions will be precised later.

Since $\overline{U}(X_t, Y_t, V_t)$ is continuous and tends to ∞ as (X_t, Y_t, V_t) tends to the boundary of $\mathbb{R}^3_+$, it admits a lower bound M^* and achieves this lower bound at $\mathbb{R}^3_+$. Therefore, we construct the non-negative function

$$\overline{U}_m(X_t, Y_t, V_t) = \overline{U}(X_t, Y_t, V_t) - M^*. \tag{4.6}$$

By simple computations (see (3.25) in subsecion 3.1), we arrive at

$$\mathcal{L}U(X_t, Y_t, V_t) \leq -2\lambda I \left(1 - \frac{1}{\mathcal{R}_0^s}\right) + \frac{\beta\lambda I^2}{\alpha_0\delta} V_t \tag{4.7}$$

where $I = \delta\left(\delta + \frac{\sigma_1^2}{2}\right)^{-1}$. In addition, by using $f(X_t, V_t) \leq \frac{\beta X_t}{\alpha_0}$, we can get

$$\mathcal{L}U_2(X_t, Y_t, V_t) \leq -\frac{\lambda}{X_t} - \eta\frac{V_t}{Y_t} + \frac{\beta}{\alpha_0}V_t + \delta + \gamma + \frac{\sigma_1^2}{2} + \frac{\sigma_2^2}{2}. \tag{4.8}$$

Similarly, by using Cauchy-Schwartz inequality, we arrive at

$$\begin{aligned}\mathcal{L}U_3(X_t, Y_t, V_t) \leq & \left(X_t + Y_t + \frac{\gamma}{k}V_t\right)^m \left(\lambda - \delta X_t + \eta V_t - \frac{\gamma(\mu+\eta)}{k}V_t\right) \\ & + \frac{3m}{2}\left(X_t + Y_t + \frac{\gamma}{k}V_t\right)^{m-1}\left(\sigma_1^2 X_t^2 + \sigma_2^2 Y_t^2 + \left(\frac{\gamma}{k}\right)^2 \sigma_3^2 V_t^2\right).\end{aligned} \tag{4.9}$$

We choose m sufficiently small so that the following inequality holds,

$$m\left(\sigma_1^2 \vee \sigma_2^2 \vee \left(\frac{\gamma}{k}\right)^2 \sigma_3^2\right) \leq \min\left(\delta, \eta, \frac{\gamma(\mu+\eta)}{k}\right) \tag{4.10}$$

Then, we get

$$\mathcal{L}U_3(X_t, Y_t, V_t) \leq -m\delta X_t^{m+1} - \frac{\eta}{2}Y_t^{m+1} - \frac{\mu+\eta}{2}\left(\frac{\gamma}{k}\right)^{m+1} V_t^{m+1} + \phi \tag{4.11}$$

where

$$\begin{aligned}\phi := \sup_{(X_tY_t,V_t)\in\mathbb{R}_+^3} \Bigg\{ & -\delta(1-m)X_t^{m+1} - \frac{\eta}{2}Y_t^{m+1} - \frac{\mu+\eta}{2}\left(\frac{\gamma}{k}\right)^{m+1} V_t^{m+1} \\ & + \frac{3m}{2}\left(\sigma_1^2 \vee \sigma_2^2 \vee \left(\frac{\gamma}{k}\right)^2 \sigma_3^2\right)\left(X_t + Y_t + \frac{\gamma}{k}V_t\right)^{m-1}\left(X_t^2 + Y_t^2 + V_t^2\right) \\ & + (\lambda + 2\eta V_t)\left(X_t + Y_t + \frac{\gamma}{k}V_t\right)^m \Bigg\}.\end{aligned}$$

Therefore

$$\begin{aligned}\mathcal{L}U_m(X_t, Y_t, V_t) \leq & -2M\lambda I\left(1 - \frac{1}{\mathcal{R}_0^s}\right) + M\frac{\beta\lambda I^2}{\alpha_0\delta}V_t - \frac{\lambda}{X_t} - \eta\frac{V_t}{Y_t} \\ & - m\delta X_t^{m+1} - \frac{\eta}{2}Y_t^{m+1} - \frac{\mu+\eta}{4}\left(\frac{\gamma}{k}\right)^{m+1} V_t^{m+1} + \Phi(m)\end{aligned}$$

where

$$\Phi := \sup_{V_t\in\mathbb{R}_+}\left\{-\frac{\mu+\eta}{4}\left(\frac{\gamma}{k}\right)^{m+1} V_t^{m+1} + \frac{\beta}{\alpha_0}V_t + \delta + \gamma + \frac{\sigma_1^2}{2} + \frac{\sigma_2^2}{2} + \phi\right\}.$$

Let ε_i, $i = 1, 2, 3$ be positive reals and define the closed set D as follows:

$$D = \left\{(X_t, Y_t, V_t) \in \mathbb{R}_+^3 \,:\, \varepsilon_1 \leq X_t \leq \frac{1}{\varepsilon_1},\, \varepsilon_2 \leq Y_t \leq \frac{1}{\varepsilon_2}\, and\, \varepsilon_3 \leq V_t \leq \frac{1}{\varepsilon_3}\right\}.$$

We divide $\mathbb{R}^3_+ \setminus D$ into six domains such that $\mathbb{R}^3_+ \setminus D = \bigcup_{i=1}^3 D_i$ where

$$\begin{aligned}
D_1 &= \{(X_t, Y_t, V_t) \in \mathbb{R}^3_+ \,:\, X_t < \varepsilon_1\},\\
D_2 &= \{(X_t, Y_t, V_t) \in \mathbb{R}^3_+ \,:\, X_t > \frac{1}{\varepsilon_1}\},\\
D_3 &= \{(X_t, Y_t, V_t) \in \mathbb{R}^3_+ \,:\, Y_t < \varepsilon_2\},\\
D_4 &= \{(X_t, Y_t, V_t) \in \mathbb{R}^3_+ \,:\, Y_t > \frac{1}{\varepsilon_2}\},\\
D_5 &= \{(X_t, Y_t, V_t) \in \mathbb{R}^3_+ \,:\, X_t \geq \varepsilon_1, Y_t \geq \varepsilon_2, V_t < \varepsilon_3\},\\
D_6 &= \{(X_t, Y_t, V_t) \in \mathbb{R}^3_+ \,:\, V_t > \frac{1}{\varepsilon_3}\}.
\end{aligned}$$

In order to verify the hypothesis (H_2), we need to show that $\mathcal{L}U(X_t, Y_t, V_t)$ is negative at any vector in $\mathbb{R}^3_+ \setminus D$. To do this, we proceed in treatment of six cases by choosing (X_t, Y_t, V_t) in the above domains.

Case 1: Let $(X_t, Y_t, V_t) \in D_1$ and $\Phi_1 = \sup\limits_{V_t \in \mathbb{R}_+} \left\{ M\frac{\beta\lambda I^2}{\alpha_0\delta} V_t - \frac{\mu+\eta}{4}\left(\frac{\gamma}{k}V_t\right)^{m+1} + \Phi \right\}$. By using the fact that $\frac{\lambda}{X_t} > \frac{\lambda}{\varepsilon_1}$, $\mathcal{R}_0^s > 1$ and choosing ε_1 sufficiently small, we arrive at

$$\mathcal{L}U_m(X_t, Y_t, V_t) \leq -2M\lambda I\left(1 - \frac{1}{\mathcal{R}_0^s}\right) - \frac{\lambda}{X_t} + \Phi_1 < 0. \tag{4.12}$$

Case 2: Let $(X_t, Y_t, V_t) \in D_2$. Then, by using $-m\delta X_t^{m+1} < -\frac{m\delta}{\varepsilon_1^{m+1}}$, $\mathcal{R}_0^s > 1$ and choosing ε_1 sufficiently small, we get

$$\begin{aligned}
\mathcal{L}U_m(X_t, Y_t, V_t) &\leq -2M\lambda I\left(1 - \frac{1}{\mathcal{R}_0^s}\right) - m\delta X_t^{m+1} + \Phi_1\\
&\leq -2M\lambda I\left(1 - \frac{1}{\mathcal{R}_0^s}\right) - \frac{m\delta}{\varepsilon_1^{m+1}} + \Phi_1 < 0.
\end{aligned} \tag{4.13}$$

Case 3: Let $(X_t, Y_t, V_t) \in D_3$. Then, by using $\mathcal{R}_0^s > 1$, there exists a positive constant $M > 0$ sufficiently large such that

$$\mathcal{L}U_m(X_t, Y_t, V_t) \leq -2M\lambda I\left(1 - \frac{1}{\mathcal{R}_0^s}\right) + \Phi_1 < 0. \tag{4.14}$$

Case 4: Let $(X_t, Y_t, V_t) \in D_4$. Since $-Y_t^{m+1} < -\left(\frac{1}{\varepsilon_2}\right)^{m+1}$, we choose ε_2 sufficiently small such that

$$\mathcal{L}U_m(X_t, Y_t, V_t) \leq -2M\lambda I\left(1 - \frac{1}{\mathcal{R}_0^s}\right) - \frac{\eta}{2}\left(\frac{1}{\varepsilon_2}\right)^{m+1} + \Phi_1 < 0. \tag{4.15}$$

Case 5: Similarly to the 'case 3', if $(X_t, Y_t, V_t) \in D_5$, then

$$\mathcal{L}U_m(X_t, Y_t, V_t) \leq -2M\lambda I\left(1 - \frac{1}{\mathcal{R}_0^s}\right) + +M\frac{\beta\lambda I^2}{\alpha_0\delta}\varepsilon_3 + \Phi_1 < 0. \tag{4.16}$$

Case 6: Let $(X_t, Y_t, V_t) \in D_4$. Then, by using $\mathcal{R}_0^s > 1$, there exists a positive constant $M > 0$ sufficiently large such that

$$\mathcal{L}U_m(X_t, Y_t, V_t) \leq -2M\lambda I\left(1 - \frac{1}{\mathcal{R}_0^s}\right) + \Phi_1 < 0. \tag{4.17}$$

With the above choices of M, ε_1 and ε_2 and for a given $\varepsilon_3 < 1$, we arrive to for any $(X_t, Y_t, V_t)\mathbb{R}^3_+ \setminus D$

$$\mathcal{L}U_m(X_t, Y_t, V_t) < 0. \tag{4.18}$$

This complete the proof of Theorem 5.

5 Numerical simulations

In order to understand a complex dynamic behavior of stochastic model (1.5) and to validate our theoretical results, we present the numerical simulations for different parameters. Using FORTRAN language, we numerically solve the stochastic differential equations of (1.5) by means of Milstein's scheme [2], for the parameter values and the initial conditions (X_0, Y_0, V_0) mentioned earlier.

5.1 Test case 1: $\sigma_1 = \sigma_2 = 0$ and $\sigma_3 = 0.05$

Firstly, we use the following parameter values mentioned in table 1.

λ	δ	k	γ	μ	η	β	α_0	α_1	α_2	α_3
10	0.014	0.5	0.27	3	0	0.000024	1	0.1	0.01	0.00001

Table 1: Parameter values of our stochastic model used in Figures 1 and 2.

Using the above parameters, we easily verify the different conditions of [4, Theorem 3.1]. We deduce that the solution of stochastic model converges to E_{eq} which is globally asymptotically stable. In this case, the virus is cleared and the infection dies out. Figure 1 demonstrates the above analysis. We retrieve the numerical result presented in [4] for the concentrations of infected cells X_t, infected cells Y_t and free virus cells V_t.

Secondly, we modify the parameter value $\beta = 0.0012$ and we keep the same parameter values in Table 1. Here, the basic reproduction number (see [4]) is $R_0 = 17.5341 > 1$ and the solution of stochastic model in [4] (with $\sigma_1 = \sigma_2 = 0$ and $\sigma_3 = 0.05$) fluctuates around the chronic infection equilibrium E^{eq} with coordinates $(85.080, 32.657, 13060 \times 10^3)$. Figure 2 illustrates the above analysis.

5.2 Test case 2: $\sigma_1 = 0.035, \sigma_2 = 0.04$ and $\sigma_3 = 0.05$

In this case, we will be interested in our stochastic model in order to illustrates the theoretical analysis. Firstly, we choose the initial value $(X_0, Y_0, V_0) = (0.5, 0.15, 0.1)$ and consider the parameter values tabulated in Table 2.

λ	δ	k	γ	μ	η	β	α_0	α_1	α_2	α_3
7.35	0.017	0.5	0.27	3.025	0.85	0.0024	1	0.1	0.01	0.00001

Table 2: Parameter values of our stochastic model used in Figure 3.

By using the above parameters, we can easily verify that the criteria conditions under which the viral infection dies out exponentially, see Theorem 2.

$$\mathcal{R}^0 = \frac{\eta}{(\mu+\eta)\gamma} = 0.8124 < 1 \tag{5.1}$$

and

$$\min(\gamma, \mu+\eta)\left(\sqrt{R_0} - 1\right) + \frac{(2\delta + \sigma_1^2)(\mu+\eta)\beta\sqrt{R_0}}{2\lambda\alpha_0\gamma} = -0.0266 < 0. \tag{5.2}$$

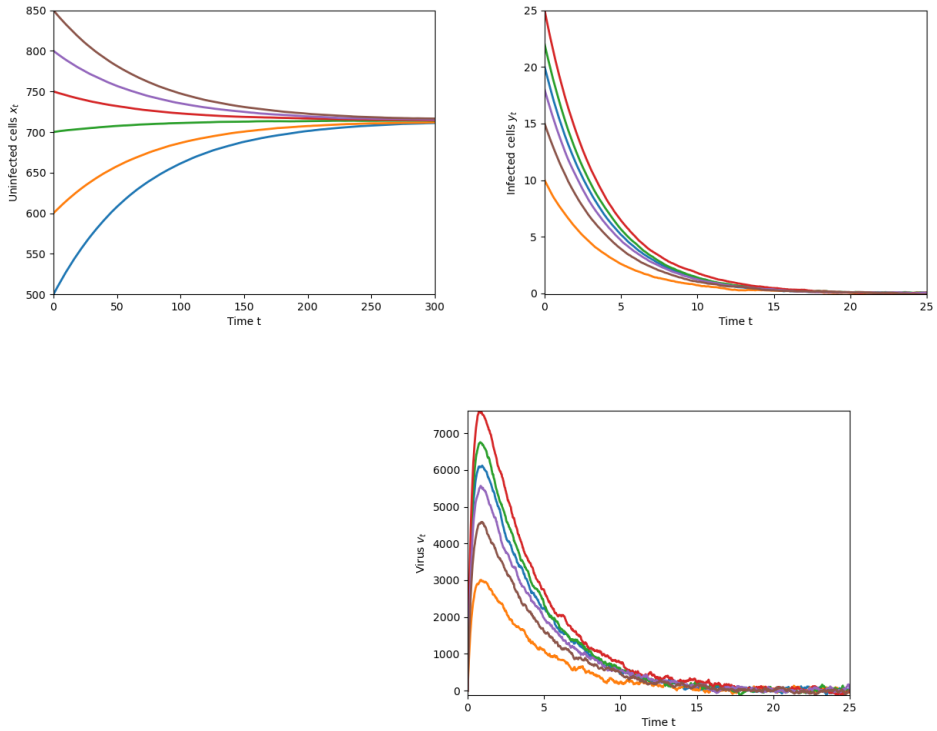

Figure 1: Extinction of viral infection when $R_0 < 1$ with parameters given in Table 1.

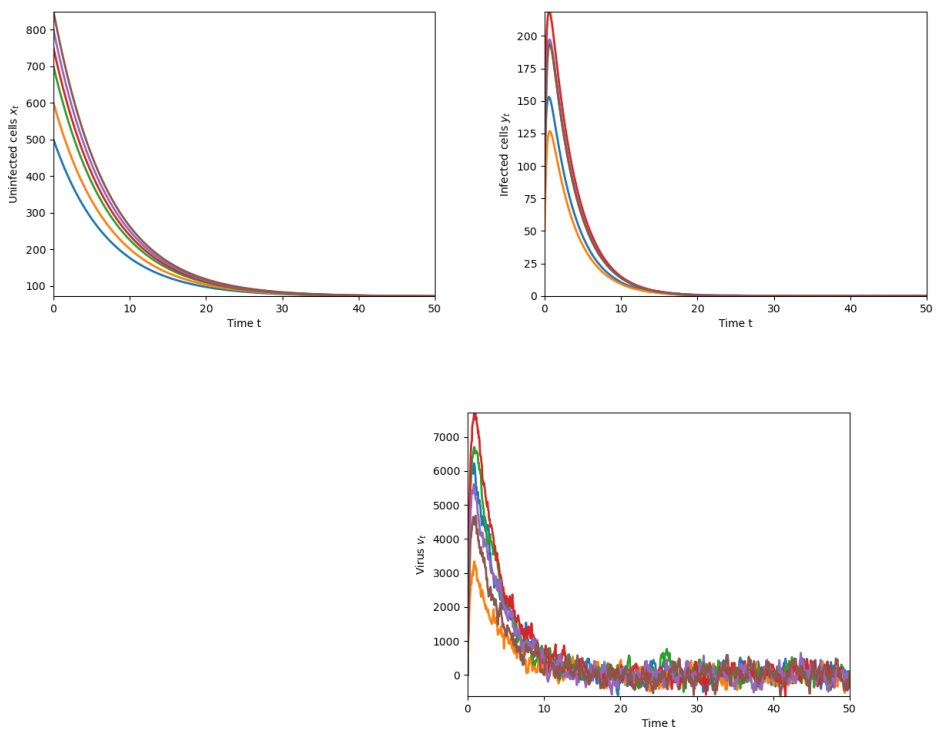

Figure 2: Extinction of viral infection when $R_0 > 1$ with parameters given in Table 1.

In this case, the virus is cleared and the infection dies out exponentially. Figure 3 illustrates this analysis.

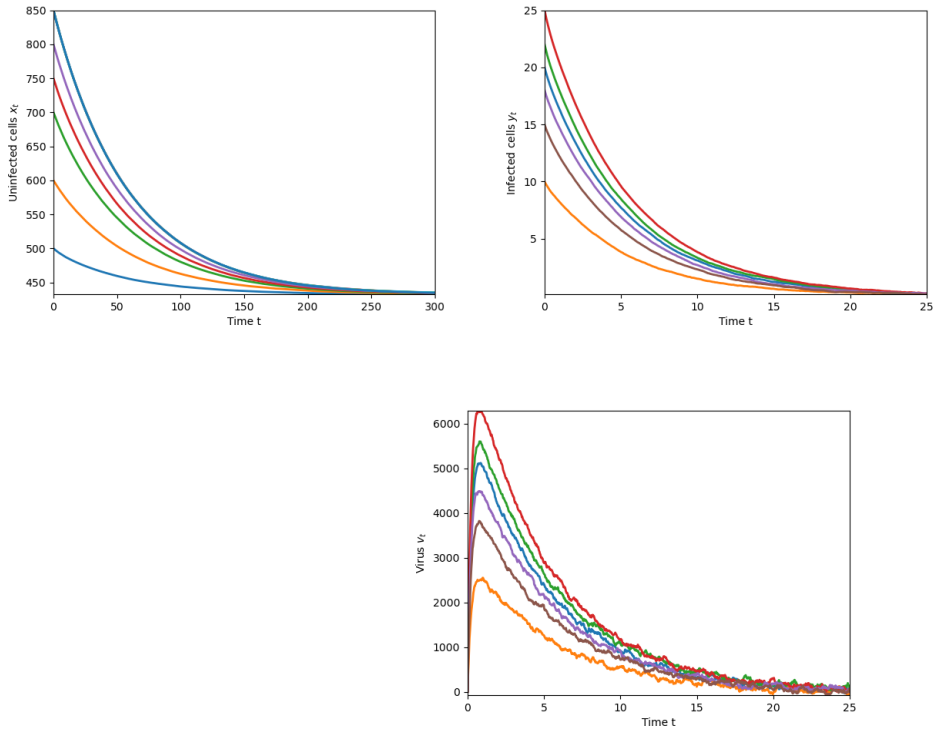

Figure 3: Extinction of viral infection when $R_0 > 1$ with parameters given in Table 2.

Secondly, we use the parameters values cited in Table 3 in order to illustrate the persistence and prevalent of the viral infection.

λ	δ	k	γ	μ	η	β	α_0	α_1	α_2	α_3
7.35	0.1	0.5	0.35	1.29	0.9	0.0037	1	0.1	0.01	0.00001

Table 3: Parameter values of our stochastic model used in Figures 4 and 5.

Using the above parameters, the conditions of Theorem 3 under which the viral infection is permanent are satisfied.

$$\mathcal{R}_0^s = \frac{2k\eta\delta}{\delta\left(\gamma + \frac{\sigma_2^2}{2}\right)\left(\mu + \eta + \frac{\sigma_3^2}{2}\right) + k\eta\left(\delta + \frac{\sigma_1^2}{2}\right)} = 1.07577 > 1. \tag{5.3}$$

In other words, the viral infection should be prevalent with probability one. Figure 4 confirms the behaviour of the virus with parameters cited in Table 3.

By sampling 20 000 stochastic paths of the solution to our stochastic viral infection model and keeping the same parameter set presented in Table 3, the stationary probability density functions of infected and removed individuals at different times are shown in Figure 5 respectively. We see that they are close to each other at $t = 2000$ and $t = 3000$, which is confirmed by Theorem 5.

6 Conclusion

The main objective of this paper consisted in proposing and analyzing a stochastic viral infection model with general functional response. As the first step, the existence and uniqueness of a global

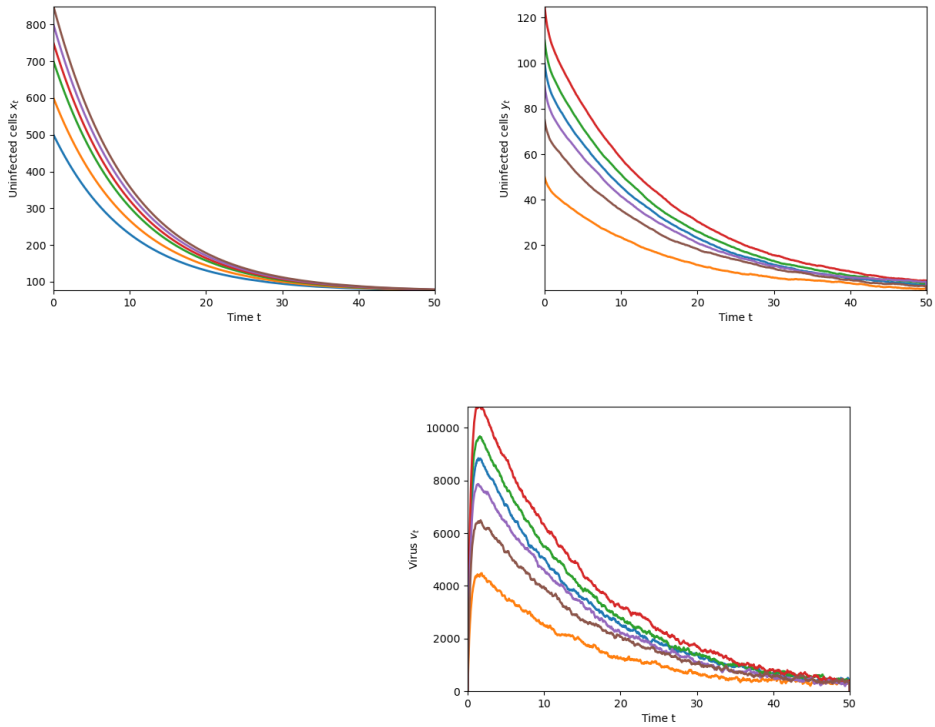

Figure 4: Extinction of viral infection when $R_0^s > 1$ with parameters given in Table 3.

positive solution is shown. Some sufficient conditions are obtained to promise the extinction of the viral infection. $R_0^s > 0$ is an adequate condition to the permanence of the disease as well as the existence of unique stationary distribution to our model. All simulations results support and validate. Future works are intended to be done in this field, such as the study of the behaviour of our model in the case where $0 \leq R_0 \leq 1 \leq R_0^s$, and the use of a second Backward stochastic differential equation in order to analyze and understand the viral infection.

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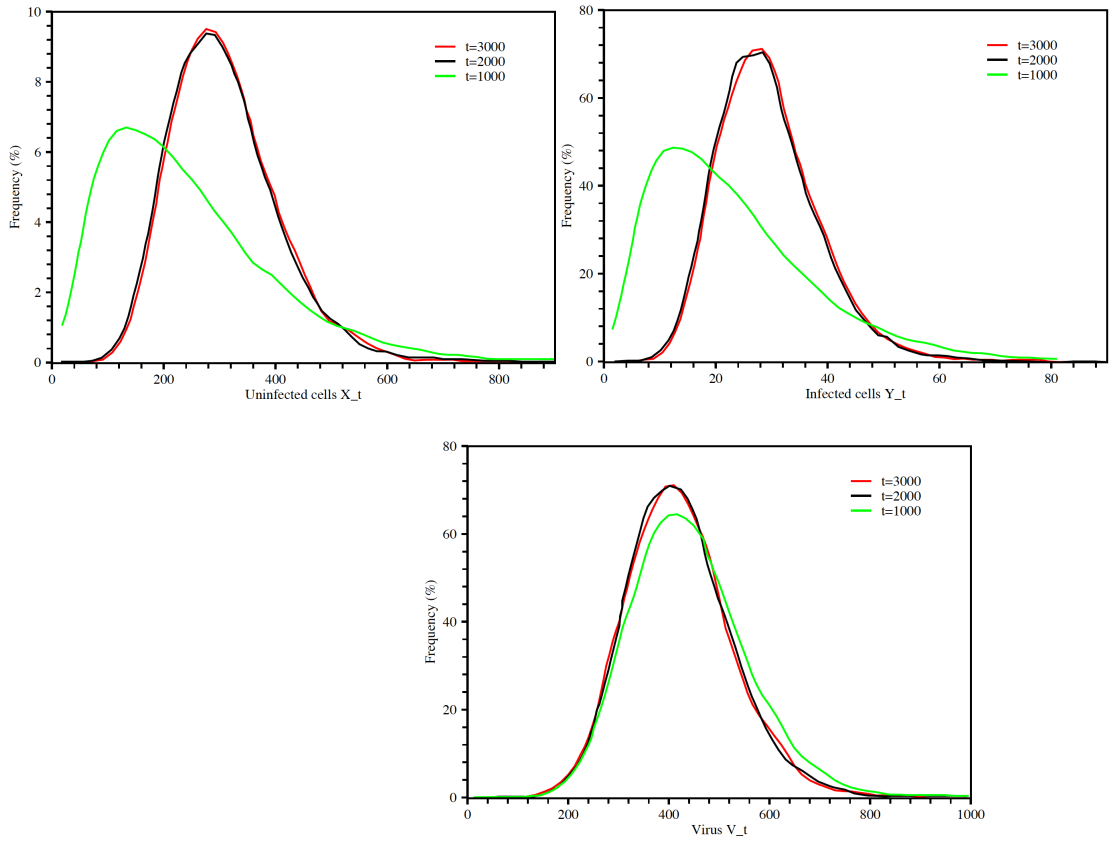

Figure 5: Kernel density functions of infected cells, uninfected cells and free virus cells at different time with parameters given in Table 3.

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